

# Receipt Accounting and Horizon Composition for Anonymous-DHT Budgets:

A Rulebook for Repeats, Retries, and State (Series Synthesis)

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## Abstract

This archive publishes anonymity budgets as small, replayable receipt line items. A recurring failure mode in the literature and in deployments is *silent horizon drift*: a proof that is per-lookup is later quoted as “per day” or “for  $Q$  lookups”, even though retries, caches, and adaptive routing change what composes. This note is the series’ compact rulebook: it states what a receipt must declare before it can honestly claim horizon- $n$  composition, and it points to the owning papers for proofs. The goal is editorial: keep mechanism papers short by making “how budgets compose across time and state” a single citable module.

**Scope.** This note concerns *repetition*: how per-lookup budgets become horizon budgets under retries, windowing, and state evolution. It does not introduce new endpoint metrics; it only standardizes when a receipt may claim “ $n$ -use” anonymity.

**Imports.** Sequential/adaptive composition for odds inflation and  $(\varepsilon, \delta)$  checkpoints is owned by Anonymity A.[4] Threat-window declarations (unit, window, reset semantics) are owned by Synthesis 4.[3] Receipt line-item schemas (including plan replay hooks) are owned by Synthesis 12.[2] Probabilistic separation adjuncts for primary-path guards (a  $(0, \rho)$  checkpoint) are owned by Synthesis 23.[16] State evolution as a witness surface is owned by the AnonDHT State series.[10]

**Exports.** We export (i) a receipt-facing definition of a *horizon claim*, (ii) a small set of conservative editorial rules (when to state “ $Q\times$ ” and when not to), and (iii) a checklist connecting horizon claims to replay plans (Operational series) and state declarations (State series).

## 1 Horizon claims are objects, not prose

Receipts in this archive are intentionally *source-only*: they do not ship full tooling and proofs. Therefore, when a receipt claims a horizon- $n$  budget, it must expose enough structure for an auditor to replay the accountant. The key principle is: *publish what you account, and account for what you publish*.

**Definition 1** (Horizon claim (receipt-facing)). *Fix a threat-window declaration (TW) naming the repetition unit (per lookup, per epoch, per day) and reset semantics.[3] A horizon claim for a receipt line item consists of:*

1. a **unit** and **window binding** (TW-ID),

2. a **per-unit budget**  $b$  (e.g., per-lookup MaxL / odds-inflation cap),
3. an **accountant**  $A$  that maps  $(b, n)$  to a cumulative budget  $b_{\text{tot}}$ , and
4. a **replay hook** committing to how  $A$  is recomputed (`plan_id` and digest-bound `plan_spec_id`).[2, 9]

The published scalar `budget_value` is then interpreted as  $b_{\text{tot}}$  for the declared unit and window.

**Remark 1** (How a horizon claim appears in a line item). The receipt tuple schema (Synthesis 12) already has the needed hooks.[2] A horizon claim is realized by: (i) binding the unit and reset semantics via `tw_id`, (ii) declaring the usage assumption (often `usage=Q` or a rate-limit clause), and (iii) committing the accountant and its evidence through the digest-bound `replay_hook` (`plan_id`, `plan_spec_id`).[9] For Regime II, the plan spec should in particular identify the conditioning interface and per-step evidence objects (Synthesis 21), so an auditor can replay the conditional-step inequality rather than assume independence.[11]

**Remark 2** (Why the replay hook matters). Without a digest-bound plan spec, two releases can both say “horizon- $n$ ” but silently use different accounting rules (e.g., naive  $n\varepsilon$  vs checkpoint composition; or different retry policies), making public comparisons meaningless. Operational A/B and the worked example show what the replay plan looks like in practice.[5, 6, 7]

**Units.** Throughout, we measure  $\varepsilon$  in *bits* so that  $2^\varepsilon$  is the multiplicative Bayes-factor / odds-inflation cap. (This matches the series convention used by Anonymity A and the endpoint bridge.)[4]

**Proposition 1** (Adaptive sequential composition for one-vs-mixture budgets). Let  $K$  be the hidden key/identity and let  $X_{1:n}$  be an interactive sequence of published witnesses (possibly chosen adaptively from past witnesses and state). Write  $P_k$  for the law given  $K = k$  and  $P = \sum_j \pi(j)P_j$  for the mixture under a fixed prior  $\pi$ . Assume that for each step  $t$  and each transcript prefix  $x_{<t}$  with  $P(x_{<t}) > 0$ ,

$$P_k(X_t \in E \mid x_{<t}) \leq 2^{\varepsilon_t} P(X_t \in E \mid x_{<t}) + \delta_t \quad \text{for all measurable } E \text{ and all } k.$$

Then the joint transcript satisfies the composed one-vs-mixture bound

$$P_k(A) \leq 2^{\sum_{t=1}^n \varepsilon_t} P(A) + \sum_{t=1}^n \delta_t 2^{\sum_{j<t} \varepsilon_j} \quad \text{for all events } A.$$

In particular, when all  $\delta_t = 0$  the realized odds inflation (log Bayes factor) composes additively:  $\ell(X_{1:n}) \leq \sum_t \varepsilon_t$ .

**Remark 3** (Ownership of proofs). The proof is the same hockey-stick / odds-inflation sequential composition argument used in Anonymity A.[4] This note records the receipt-facing condition: you may claim a horizon- $n$  budget without independence if you commit (via the replay hook) to a plan that certifies the per-step conditional bound above.

**Remark 4** (Separation-failure adjuncts compose as  $\delta$ ). If a primary-path surface is guarded by a separation manifest (Synthesis 22) and publishes a separation-failure upper bound  $\rho$  (Synthesis 23), then the primary claim contributes a checkpoint  $(0, \rho)$ . Receipt-side horizon accounting treats this exactly like any other  $\delta$  term: it does not add to the  $\varepsilon$  sum and it adds at most  $\rho$  to the final additive slack (or appears as one step with  $\delta_t = \rho$  in Proposition 1).[15, 16] In particular, a horizon- $n$  claim that composes per-step budgets  $\{(\varepsilon_t, \delta_t)\}$  and carries one separation guard becomes  $(\sum_t \varepsilon_t, \rho + \sum_t \delta_t 2^{\sum_{j<t} \varepsilon_j})$  under the one-vs-mixture semantics.

## 2 Three composition regimes and what receipts may say

This rulebook separates three common regimes. A receipt should *name* which regime it is in; otherwise it should publish only per-unit budgets.

### 2.1 Regime I: independent repeats (rare, but clean)

If the repeated witnesses are conditionally independent given the secret (or satisfy the independence conditions declared by the tiered-observation interface), then additive composition is often justified.[1, 8] This is the cleanest case, but it is fragile for DHTs because routing tables, caches, and adversarial adaptivity create shared state.

### 2.2 Regime II: adaptive sequential composition (default for retries)

If the mechanism is used interactively, with later choices depending on earlier prefixes, then independence should not be assumed. The receipt-grade condition is a *per-step conditional* one-vs-mixture bound; under that condition, horizon composition follows by adaptive sequential composition (proposition 1). (certificate interface: Synthesis 21).[11] Operational A/B show how to force retries and fallback decisions into auditable schedules so the per-step claim is meaningful, and Anonymity A owns the underlying composition proof.[5, 6, 4]

### 2.3 Regime III: stateful repeats (caches, scoring, routing churn)

For anonymous DHTs, the secret can influence persistent state (routing-table updates, cache hits, score changes), and that state can later influence which witnesses are exposed. In this regime, a naive “ $Q \times$ ” claim is usually false unless the state surface is explicitly budgeted. The State series treats state evolution as a first-class witness tier and exports the declarations needed for honest accounting.[10]

**Remark 5** (Editorial rule: no horizon- $Q$  without an explicit regime). *A receipt may publish a horizon- $Q$  (or “ $Q \times$ ”) total only if it declares which regime it is using: (i) an explicit conditional-independence declaration justifying product-form repeats (Regime I), (ii) a per-step conditional budget that can be replayed (Regime II; proposition 1), or (iii) a digest-bound state declaration (*state\_decl\_id*) that enlarges the witness family so the per-step conditional bound is actually true in the presence of caches, routing churn, scoring, or other shared state (Regime III).[9, 10] Otherwise, the receipt should publish per-unit budgets only and leave horizon claims to a later, explicitly certified accountant.*

## 3 Research warranted: dependence-aware equalization without bloat

The series currently takes a conservative editorial stance: if state is not controlled and declared, do not claim  $Q \times$  composition. Two research directions can reduce conservatism while staying receipt-grade:

1. **Contraction-based accounting.** If secret influence contracts over time (e.g., via random-walk delegation, cover mixing, or a spectral gap), then horizon growth can be a discounted sum rather than linear. The backbone spectral paper provides the overlay-specific distinguishability identities used by several budget papers.[12] A receipt-grade interface would expose the contraction factor as a knob with a dual certificate (mirroring the TV dual-certificate pattern).[14]

2. **Stop-time normal forms for adaptive deadlines.** Retries and deadlines often induce stopping-time effects. The stop-time padding addendum provides normal forms (deadline tails, mixture separations) that can be used to keep accounting honest under optional stopping.[13] A compact “optional-stopping accountant” could then be referenced by both runtime receipts (Operational series) and evaluation receipts (Evaluation 1).

These are intentionally stated as interface targets: once the owning mechanism paper exports a knob+certificate for contraction or stop-time effects, this note becomes the canonical place to say how it composes.

## 4 Checklist for authors and auditors

A paper (or receipt) that wants to state a horizon budget should satisfy:

1. **TW declared:** unit, window, reset semantics (Synthesis 4).
2. **Per-unit budget exported:** odds inflation / MaxL / hockey-stick (mechanism paper + endpoint bridge).
3. **Accountant named:** which theorem/recipe is being used (Anonymity A; State series; Boss Fight budgets).
4. **Replay hook bound:** `plan_id` + `plan_spec_id` for auditors (Synthesis 12; Operational B; worked example).
5. **State scope honest:** either independence is justified, or `state_decl_id` binds which state surfaces are included (State series; Synthesis 18).

## References

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